

Conserved amount, negotiable placement: prompting moves one architecture’s complexity distribution and not the other’s

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Abstract

Tesler’s conservation law holds that a problem carries an irreducible amount of complexity, and Brooks separates that essential part from the accidental part a solution adds. If the essential part is fixed, then architecture is a question of placement rather than quantity. This paper tests both halves of that claim on a controlled experiment in which an AI coding agent implements sixty accumulating change requests against a single REST endpoint, ten independent runs per cell, with the model held fixed. The independent variables are the architecture (a mutative SPRING controller against an additive OFFICEFLOOR composed pipeline) and the instruction the agent receives, which is one of four: a bare change request, a plain language request for cohesion, a tool that grades each change and asks for a refactor, and the structural cost function itself pasted into the prompt.

Across all eight cells the amount of complexity in the finished system varies by 7% on total Halstead volume and 15% on total cyclomatic complexity, and the complexity *added* by the sixty rules varies by 21% to 29% within an architecture. Placement does not behave this way. This paper defines a per metric *intervention spread*, the range across the four conditions divided by their mean, and reports the ratio of the two architectures’ spreads. On amount metrics that ratio is near 1 and no confidence interval excludes 1: both architectures are equally immovable. On eleven placement metrics it runs from 2.9 to 10.0, and every interval lies entirely above 1, entirely because the mutative architecture moves. Under its strongest intervention the mutative arm arrives at the placement the additive arm occupies with no intervention at all. The price is reported in full. Correctness fell under every intervention in both architectures, three arrangement metrics fail to separate the arms, a location-blind erosion measure borrowed from prior work ranks the additive architecture worse in three of four conditions, and the placement result predicts none of the outcome measures: the concentrated codebase was recovered better by a cold reader in all four conditions and was cheaper to change in three of four. The contribution is a measurable property of an architecture, not evidence that either architecture is better.

1 Introduction

Larry Tesler’s law of conservation of complexity holds that a problem carries an irreducible amount of complexity, and that design moves it rather than removing it. Tesler’s own formulation, from around 1984, puts the emphasis on placement rather than quantity: “Every application has an inherent amount of irreducible complexity. The only question is: who will have to deal with it, the user, the application developer, or the platform developer?” [21, 11]. This paper asks Tesler’s question one level further in, of two architectures rather than of three audiences. Brooks draws the same line from the other direction, separating the *essential* complexity that belongs to the problem from the *accidental* complexity that a particular solution adds [2]. Both claims are widely quoted and rarely measured, because measuring them needs

a setting in which the problem is genuinely held fixed while the solution is genuinely allowed to vary.

An AI coding agent working through a fixed list of change requests is such a setting. The task is identical by construction. The model is identical. What can be varied is the architecture the change lands in, and the instruction the agent is given before it writes.

This paper reports a 2×4 study. Two architectures receive sixty accumulating change requests each, under four instruction conditions, ten independent runs per cell: 4,800 agent implementations in total.

The usual question to ask of such a design is which architecture ends up with the better structure. This paper asks a prior one. If the amount of complexity is fixed by the problem, then every architectural argument is an argument about arrangement, and the interesting property of an architecture is not the arrangement it happens to have but *how far that arrangement can be moved by anything other than the architecture itself*. An architecture whose structure is set by the prompt, the review or the dashboard is only as durable as the discipline applying them.

This paper contributes:

1. A four way measurement of Tesler conservation under a fixed task, with four independent operationalisations of “amount”, reported both as finished totals and as the complexity the change requests added.
2. A statistic, the *intervention spread*, that separates metrics measuring amount from metrics measuring arrangement by how far an intervention can move them, and a ratio between architectures that is approximately 1 on the former and 3 to 10 on the latter.
3. The finding that the mutative architecture’s placement under its strongest intervention converges on the additive architecture’s placement under no intervention.
4. A full accounting of what the interventions cost, including a correctness collapse in both architectures, three arrangement metrics that fail to separate the architectures at all, and a borrowed degradation measure that ranks the result the other way round.

2 Background and related work

Complexity as a conserved quantity. Tesler’s law is normally invoked in interface design, where the question is whether the user or the program absorbs a step [21, 11]. Its software engineering form is Brooks’ essence and accident distinction [2]. Neither is usually stated in a way that a measurement could falsify, which is what this study tries to supply.

Measuring amount. Four operationalisations are used, resting on different theories, so that agreement between them is evidence rather than an artefact of one definition: McCabe’s cyclo-matic complexity over control flow [10], Halstead volume over program vocabulary [6], weighted methods per class from the Chidamber and Kemerer suite [4], and PMD’s cognitive complexity over nesting.

Measuring arrangement. Concentration is measured with the Herfindahl-Hirschman index and the Gini coefficient over per file and per class complexity shares, which answer “how unequally is this spread” while staying silent on the total. Change is measured with Hassan’s change entropy [7], applied cumulatively over the whole run rather than per commit. Coupling is measured with CK’s coupling between objects [4] and with MacCormack’s propagation cost [9]. The Maintainability Index [12] is computed per file and averaged. Cohesion is measured with LCOM [4, 8].

Measurement under optimisation pressure. Goodhart’s law, in Strathern’s formulation, holds that a measure which becomes a target ceases to be a good measure [5, 20]. Two of the four conditions apply deliberate optimisation pressure, and one applies it to a disclosed formula. Which metrics survived that pressure and which did not is therefore reported, and the optimised score itself is excluded from every claim.

Agentic degradation benchmarks. Two recent benchmarks shape this design. The verbosity measure, and the practice of contrasting prompt arms against a plain control, follow SlopCodeBench [13]. The strict pass criterion, under which a checkpoint counts only when every test selected so far is green, follows SWE-CI’s gate semantics [3]. SlopCodeBench’s whole-application erosion measure is also computed over these runs, and Section 5.4 reports it, because it points the other way. The harness records a wider set of measures from both benchmarks, including degradation slope, normalized change and zero regression rate; those are not used here.

3 The question and the statistic

Suppose the amount of complexity is fixed by the problem. Then two architectures that solve it differ only in arrangement, and the arrangement of any given codebase is the joint product of the architecture and everything else that shaped it: the prompt, the review, the metric on the dashboard. An architecture whose arrangement is substantially determined by those other things is one whose structure is only as durable as the discipline applying them. An architecture whose arrangement is determined by its own composition is not.

That is an empirical question, and it needs a statistic. For a metric m and an architecture a , let $\mu_c(m, a)$ be the mean over ten runs of the per run mean of m over the final phase, under condition c . The **intervention spread** is

$$S(m, a) = \frac{\max_c \mu_c(m, a) - \min_c \mu_c(m, a)}{|\mu_c(m, a)|} \quad (1)$$

where c ranges over the four conditions, and the **plasticity ratio**

$$R(m) = S(m, \text{SPRING}) / S(m, \text{OFFICEFLOOR}). \quad (2)$$

S asks how far an intervention moved an architecture on that metric, expressed relative to the metric’s own level so that metrics on different scales are comparable. R asks whether the two architectures were moved equally. Under Tesler conservation, amount metrics should have small S for both architectures, and therefore $R \approx 1$. The thesis of this paper is that placement metrics have $R \gg 1$.

Both are computed over four condition means, each itself a mean over ten runs, so both carry sampling error. An interval is attached by a **cluster bootstrap over chains**: within each of the eight cells the ten chains are resampled with replacement, the cell mean recomputed, and S and R recomputed from the resampled means, 10,000 times. The reported interval is the 2.5th to 97.5th percentile of that distribution.

The chain is the resampling unit, not the checkpoint. Checkpoints within a chain are successive states of one codebase and are strongly dependent, so treating them as independent would give intervals far too narrow. Chains are independent runs. Each cell is resampled separately, because chain k of one condition has no relationship to chain k of another.

4 Study design

Task. A fixed plan of sixty change requests, each adding or revising a business rule on one endpoint, `POST /api/owners`, of the Spring PetClinic REST application. The plan and its acceptance suite were frozen on 2026-08-08, before the earliest of the four runs reported here, so all four conditions faced an identical task.

Architectures. The *mutative* arm is an idiomatic Spring [19] `@RestController`: a rule is a statement added to a handler method. The *additive* arm is the same application built with OfficeFloor [18], a framework whose endpoints are YAML composed graphs of functions, so a rule is a new function and a new edge. Both arms implement the same sixty rules and are held to the same black box acceptance suite, which the agent never sees.

Agent. `claude-opus-4-8`, held fixed across all cells. Each change request is a fresh session with no memory of the previous one, so the agent re-reads whatever it needs from the code itself. Nothing but the code crosses between checkpoints.

Conditions. Four, each a full sweep of both architectures at ten runs each, each changing exactly one lever:

1. **just-solve.** Control. The change request and nothing else.
2. **cohesion-prompt.** The *prompt* lever. One plain language paragraph asking for well structured code. No metric is named and the experiment is not mentioned.
3. **impact-gated.** The *tool* lever. A neutral implement prompt, but a change that grades badly gets one quality gated refactor guided by the flagged locations, then is accepted whatever its grade. The gate fired on 195 of 1,200 checkpoints.
4. **formula-provided.** The *disclosure* lever. The structural cost function itself pasted into the implement prompt as the objective. Its gate fired on 3 of 1,200 checkpoints, so this condition is the prompt lever, measured.

Measurement. Every metric is recomputed from the committed source at every checkpoint by `lizard`, `PMD`, `CK` [1] and `jscpd`, never read back from anything the agent wrote. The population is production Java only. OfficeFloor’s YAML wiring is counted in a separate pool and is never folded into a Java denominator, because doing so would let a wiring based architecture dilute any per line metric.

Aggregation. A run is binned into fifths of twelve change requests. Unless stated otherwise, every reported value is the final phase, meaning change requests 49 to 60, averaged within a run first and then across the ten runs, so that a run missing a checkpoint does not get a smaller vote than a complete one. Parenthesised figures are one standard deviation across the ten runs.

Scale. $4 \text{ conditions} \times 2 \text{ architectures} \times 10 \text{ runs} \times 60 \text{ change requests} = 4,800$ agent implementations, at a total implement turn model spend of \$6,815.

5 Results

5.1 The amount is conserved

Table 1 gives the four amount operationalisations for all eight cells. They rest on different theories and different tools, and they agree. Total Halstead volume varies by 7% across all

Table 1: Amount of complexity in the finished system, final phase, mean (SD) over ten runs. “Spread” is the range over all eight cells divided by their mean.

Metric	Arch	just-solve	cohesion	gated	formula	Spread
total_cc	SPRING	652 (21)	694 (17)	646 (23)	642 (38)	15%
	OFFICEFLOOR	656 (12)	681 (28)	641 (20)	597 (19)	
halstead_volume	SPRING	215,300 (6,410)	220,800 (6,480)	209,000 (5,410)	206,100 (12,300)	7%
	OFFICEFLOOR	216,600 (1,310)	221,500 (2,110)	214,100 (3,910)	207,800 (1,690)	
ck_wmc_total	SPRING	762 (20)	826 (18)	761 (21)	750 (38)	20%
	OFFICEFLOOR	738 (12)	766 (27)	725 (21)	679 (17)	
pmd_cognitive_total	SPRING	323 (14)	293 (12)	293 (11)	283 (24)	18%
	OFFICEFLOOR	319 (17)	289 (22)	310 (23)	268 (14)	

Table 2: Complexity added by the sixty change requests, computed per run as the final phase mean minus the value at that arm’s `base_ref`, the commit every run starts from. That base is one fixed codebase per arm, identical across all four conditions, so the SD shown is the between run spread at the end. Mean (SD) over ten runs. “Spread” is within an architecture, across the four conditions.

Metric	Arch	just-solve	cohesion	gated	formula	Spread
total_cc	SPRING	248 (21)	290 (17)	242 (23)	238 (38)	21%
	OFFICEFLOOR	305 (12)	330 (28)	290 (20)	246 (19)	29%
halstead_volume	SPRING	59,021 (6,406)	64,544 (6,480)	52,700 (5,406)	49,847 (12,324)	26%
	OFFICEFLOOR	61,287 (1,305)	66,174 (2,107)	58,803 (3,915)	52,507 (1,691)	23%
ck_wmc_total	SPRING	244 (20)	308 (18)	243 (21)	232 (38)	29%
	OFFICEFLOOR	307 (12)	334 (27)	294 (21)	248 (17)	29%

eight cells, total cyclomatic complexity by 15%. No condition reached under the floor, in either architecture.

Finished totals are the quantity a team maintains, but the two applications do not start level. Each arm begins from one fixed commit, identical for all four conditions, and at that commit the mutative arm carries 404 points of cyclomatic complexity across 55 files and the additive arm 351 across 120. A matching total is therefore not automatically a matching amount of work, and Table 2 repeats the test on the complexity the sixty rules *added*.

The conservation claim survives the stricter test and picks up a qualifier. The spread widens from 7% on totals to between 21% and 29% on what was added, so the honest statement is that the amount is *steady* rather than fixed. And the additive architecture added more complexity than the mutative one in every condition. That is the price of a composed pipeline. It is a real cost, and it is visible before any argument about placement begins. The additive arm’s advantage is not that it carries less. It is that its baseline was lower and it stayed where it was put.

5.2 The placement is not

Table 3 gives the intervention spread for both architectures and the plasticity ratio between them, for eleven placement metrics and, for contrast, the amount metrics from Table 1. Figure 1 shows the same result.

The separation is clean, and it survives the interval. On the eleven placement metrics R lies between 2.9 and 10.0, and **all eleven have a 95% confidence interval lying entirely above 1**. On the six amount metrics R lies between 0.6 and 1.1 and **not one interval excludes 1**, which is the conservation result restated as a test rather than as an observation. The amount group is an internal control that behaves as it should: it was predicted to show no asymmetry, and it shows none.

Two of the eleven deserve less weight than the rest. `propagation_cost` only just clears

Table 3: Intervention spread S per architecture, and the plasticity ratio R . Final phase. A ratio near 1 means the four conditions moved both architectures equally.

Metric	$S(\text{SPRING})$	$S(\text{OFFICEFLOOR})$	R	95% CI for R
<i>Placement</i>				
cum_change_top1	110%	11%	10.0	[4.13, 30.35]
cum_change_entropy_norm	24%	3%	9.2	[6.21, 14.67]
cum_change_hhi	142%	20%	7.3	[4.17, 11.32]
ccdist_file_top1	87%	17%	5.3	[2.57, 9.23]
ck_cbo_mean	14%	3%	4.2	[2.59, 6.48]
ccdist_file_gini	34%	8%	4.1	[3.24, 5.43]
total_files	37%	9%	4.0	[2.87, 5.50]
ccdist_file_hhi	106%	28%	3.7	[2.70, 5.45]
node_cc_median	87%	24%	3.5	[2.03, 6.16]
wmcdist_class_hhi	100%	29%	3.4	[2.53, 4.98]
propagation_cost	36%	12%	2.9	[1.06, 9.27]
<i>Amount</i>				
halstead_volume	7%	6%	1.1	[0.67, 1.70]
java_loc	11%	10%	1.1	[0.73, 1.42]
total_fns	15%	16%	1.0	[0.75, 1.22]
pmd_cognitive_total	13%	17%	0.8	[0.49, 1.18]
ck_wmc_total	10%	12%	0.8	[0.60, 1.18]
total_cc	8%	13%	0.6	[0.43, 0.98]
<i>Arrangement metrics that do not separate the architectures</i>				
ck_lcom_mean	65%	47%	1.4	—
ccdist_fn_gini	15%	14%	1.1	—
cogdist_fn_gini	16%	16%	1.0	—

the line at [1.06, 9.27], and `cum_change_top1`’s interval spans [4.13, 30.35], which is the small denominator instability discussed in Section 7 made visible. The conclusion rests on neither: nine of the eleven have a lower bound above 2.

The third group is reported because it does not fit. Two distribution measures taken over individual functions rather than over files or classes, `ccdist_fn_gini` and `cogdist_fn_gini`, move by the same amount in both architectures, which suggests the inequality of complexity *within* a population of functions is architecture neutral and that the signal lives at the level where the architectural decision is actually made. `ck_lcom_mean` is the more awkward one: cohesion is what the plain-language prompt asked for by name, and it separates the arms less than almost anything else in the table. It also improved most under the condition that never mentioned it, falling from 29.6 to 17.3 in the mutative arm under the formula against 26.0 under the cohesion prompt.

5.3 The whole distribution, not the chosen eleven

Eleven metrics out of a much larger set were chosen by the author, so the obvious question is what the rest do. Figure 2 answers it for every metric the study records.

Two things follow, and they point in opposite directions. The direction of the effect is robust: 116 of 169 metrics have $R \geq 1$, only 3 fall at or below 1/2.9, and 45 reach or exceed 2.9, which is the weakest ratio the paper reports. There is no selection of eleven metrics from this set that would reverse the sign of the result.

The magnitude is not robust in the same way. The median metric gives $R = 1.32$, and the reported eleven sit in the upper tail rather than at the centre. The honest reading is that the asymmetry is pervasive and usually mild, and that it is large on the specific family of metrics

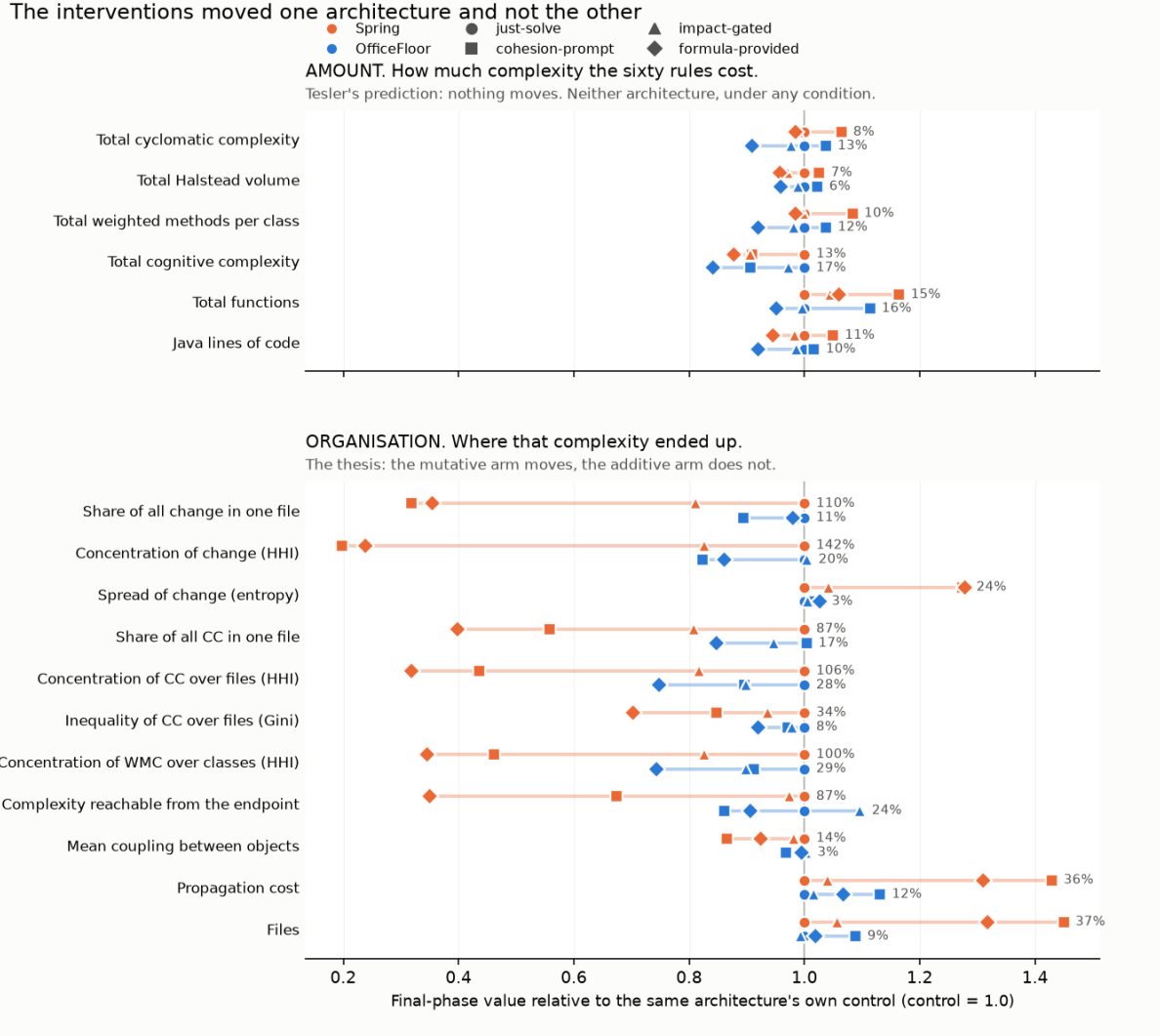


Figure 1: Every condition’s final phase value divided by the same architecture’s own control, so 1.0 means the intervention changed nothing. Top panel is amount, where neither architecture moves. Bottom panel is placement, where one does. Normalising to each architecture’s own control is deliberate, because the question is how far each moved rather than where it started; absolute levels are in Tables 1 and 4.

that measure concentration of complexity and of change. That is the claim the paper makes, but a reader should take the 2.9 to 10.0 range as describing that family and not as describing measurement of this codebase in general.

5.4 A location-blind metric points the other way

One measure from the benchmarks this design borrows from contradicts the result, and it is reported here rather than left out. SlopCodeBench’s whole-application *erosion* is the share of total complexity mass held by functions above a cyclomatic threshold. Measured over the same runs it ranks the additive arm as the more eroded one in three of the four conditions.

Condition	SPRING	OFFICEFLOOR	scored worse
just-solve	0.087	0.126	OFFICEFLOOR
cohesion-prompt	0.048	0.043	SPRING
impact-gated	0.044	0.058	OFFICEFLOOR
formula-provided	0.008	0.023	OFFICEFLOOR

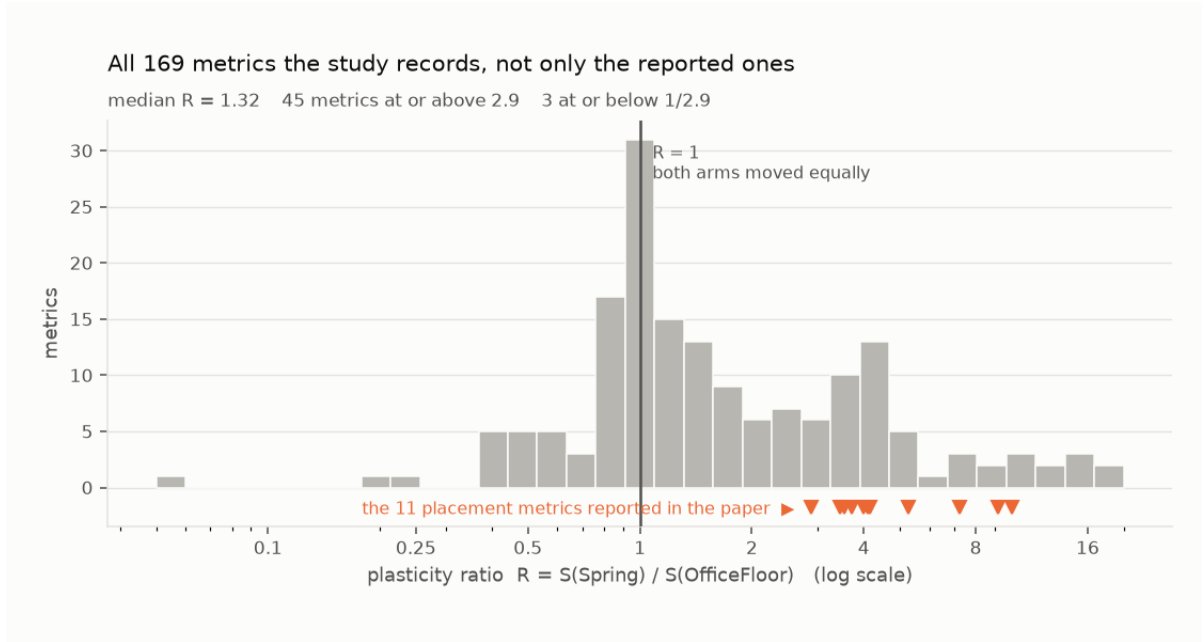


Figure 2: The plasticity ratio for all 169 metrics with a computable R , on a log scale so that halving and doubling are the same distance from parity. The markers show where the eleven reported placement metrics fall.

This is not an anomaly that the argument has to survive. It is what the argument predicts. Erosion is location-blind by construction: it cannot distinguish a cyclomatic complexity of 19 sitting in a god method from the same 19 sitting in an isolated single-responsibility algorithm. The whole claim of this paper is that measures of that kind cannot see placement, and erosion is one of them.

The mechanism is visible in the change request plan. Eight of the sixty rules concern E.164 phone normalisation, three concern Luhn checksums and two concern soundex matching. These are self-contained algorithms with genuinely high branch counts, both arms must implement them, and neither architecture makes them simpler. They therefore dominate the numerator. The count of functions over the threshold moves the same way, 3.3 in the additive arm against 1.7 in the mutative one under the control, which is what spreading the same work across more units does to a threshold count.

So a reader who prefers erosion to the placement family would reach the opposite conclusion from this data. We think that preference is mistaken, for the reason above, and Section 7 states the case a reader would have to accept to hold it. The metric is recorded by the harness for every checkpoint of every run and is in the published data.

5.5 The mutative arm converges on the additive arm’s untouched position

The interventions worked. Table 4 shows what “worked” converged on. The comparison throughout is against the additive arm under the control condition, not against either arm’s starting commit.

Coaching brings the mutative architecture to roughly the placement the additive architecture occupies with none. “Roughly” is the right word and the table is shown at full precision so it can be checked: on three of the five metrics the additive arm’s untouched control is still ahead of the mutative arm’s best, and on two the mutative arm passes it. The claim is convergence on the same neighbourhood, not that coaching closes the gap exactly.

Two caveats attach to this table and neither is small. The best of three is a maximum over conditions, which is biased upward when each cell carries run to run noise, and no correction

Table 4: Final phase placement. “Best” is whichever of the three interventions scored best on that metric for the mutative arm, and “from” names it. “OFFICEFLOOR control” is the additive arm with no intervention at all. The last column says which of those two is still ahead.

Metric	SPRING ctl	SPRING best	from	OFFICEFLOOR <i>ctl</i>	ahead
cum_change_top1	0.344	0.110	cohesion	0.106	OFFICEFLOOR
ccdist_file_top1	0.197	0.078	formula	0.086	SPRING
ccdist_file_hhi	0.0727	0.0232	formula	0.0217	OFFICEFLOOR
wmcdist_class_hhi	0.0554	0.0191	formula	0.0182	OFFICEFLOOR
cum_change_entropy_norm	0.721	0.921	formula	0.886	SPRING

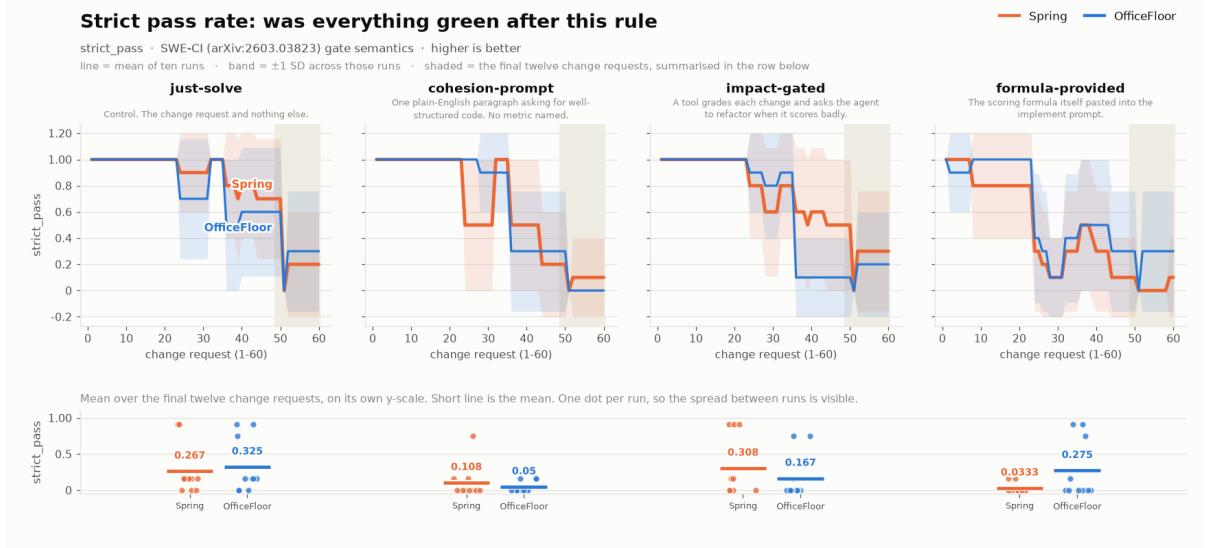


Figure 3: Strict pass rate. The four panels are the trajectory over the sixty change requests. The row beneath puts the eight final phase means side by side with one dot per run, which is where the bimodality is visible: in every cell most runs sit at 0 or 0.167 while two or three sit near 0.9, so the mean falls in a gap where almost no run lands.

is applied. And the winning condition differs by row, four from the formula and one from the cohesion prompt, so no single intervention achieves the whole of this. Read the table as showing what the best available coaching reaches, metric by metric, rather than as the profile of any one condition.

5.6 What the interventions cost

Correctness. The strict pass rate, meaning every test of every rule so far is green, fell under every intervention in both architectures. The mutative arm went from 0.267 under the control to 0.108 under the cohesion prompt and 0.033 under the formula; the additive arm went from 0.325 to 0.050 under the cohesion prompt. Both architectures are plastic here ($R = 1.1$), and structural stability did not buy safety.

These particular means need more care than the rest of the paper, because the underlying distribution is bimodal and the mean falls in a gap where almost no run lands. In every one of the eight cells most runs finish at 0 or 0.167 and two or three finish near 0.9. The mutative control mean of 0.267 is three runs at 0, five at 0.167 and two at 0.917, as Figure 3 shows. The ordering between conditions survives, because the count of runs stuck at 0 moves the same way the mean does, from 3 of 10 under the control to 8 of 10 under the formula. The level does not.

Table 5: Outcome measures, final phase, mean (SD) over ten runs. These are levels, not spreads. `probe_recall` is a read only comprehension probe: a fresh agent, given the finished codebase and none of the history, is asked to recover the rules, and higher is better. The other two are lower is better.

Metric	Arch	just-solve	cohesion	gated	formula	SPRING better in
<code>probe_recall</code>	SPRING	0.639 (0.111)	0.573 (0.040)	0.590 (0.055)	0.688 (0.055)	4 of 4
	OFFICEFLOOR	0.602 (0.025)	0.555 (0.060)	0.589 (0.071)	0.613 (0.045)	
<code>cost_usd</code>	SPRING	1.64 (0.10)	2.01 (0.18)	1.68 (0.19)	1.51 (0.13)	3 of 4
	OFFICEFLOOR	1.90 (0.16)	2.24 (0.21)	1.87 (0.20)	1.51 (0.09)	
<code>duration_api_ms</code>	SPRING	219,400	285,700	223,100	256,500	3 of 4
	OFFICEFLOOR	294,600	336,300	290,300	254,300	

Escape from the call graph. Under the formula condition the mutative arm’s count of container dispatched classes, meaning advice, aspects, filters and entity listeners, rose from 3.0 (SD 0) to 11.2 (SD 9.4) per run. That is complexity leaving the call graph rather than leaving the codebase, and it is why no single call graph scoped metric is treated as decisive here. The additive arm’s count is exactly 1.0 in all four conditions.

Duplication. Verbosity rose in the mutative arm from 0.851 to 1.002 under the cohesion prompt and 0.991 under the formula, on a codebase that did not grow. Distributing logic across more units creates the opportunity to copy between them, and it was taken.

Money. Cost per run rose under the cohesion prompt, from \$78.40 to \$96.73 in the mutative arm and \$86.51 to \$101.18 in the additive arm, and fell slightly under the formula.

5.7 What the placement result does not predict

The reason to care about placement is the belief that it governs what a team experiences: comprehension, cost, time, safety. On this study’s own outcome measures that belief is not supported, and the direction is against the additive architecture rather than merely absent.

Table 5 is the result this paper least wanted. The cold reader recovered *more* of the accumulated rules from the concentrated codebase than from the distributed one, in all four conditions. The concentrated codebase was cheaper to change in three of four and faster in three of four. Correctness, reported above, splits two and two. Under the plasticity statistic these outcome measures behave like amount rather than like placement: R is 1.9 for `probe_recall`, 0.7 for `cost_usd` and 0.9 for `cache_read_tokens`, so the interventions moved both architectures about equally on all of them.

No confident account of the comprehension result is available. One possibility is that the probe rewards locality: sixty rules in one file can be read in one pass, whereas the same rules spread over a graph of small functions require the reader to reconstruct the composition before any individual rule means anything. If that is right then concentration is bad for changing code and good for reading it, and the two are genuinely in tension rather than one being a proxy for the other. Another possibility is that the probe is simply a weak instrument. These cannot be separated here.

What this does establish is the boundary of the paper’s claim. The claim is about a structural property and its stability under instruction. It is *not* an empirical claim that the additive architecture is cheaper, faster or easier to understand, and on this evidence it is not. A reader who wants the placement result to imply a maintenance benefit will have to get that implication from somewhere other than this experiment.

6 Discussion

A Spring endpoint is a method, and a rule is a statement added to it. Nothing in the framework says where that statement goes, so each of the sixty rules is a fresh decision. Sixty decisions are sixty opportunities for the surrounding instruction to change the answer, which is what Table 3 measures: one architecture, four instructions, four materially different shapes.

An OfficeFloor endpoint is a graph of wired functions, and a rule is a new function and a new edge. The composition is the restriction. There is no version of “add a rule” that concentrates it, so a prompt asking for cohesion is asking for something already done. That is why the additive arm’s placement metrics barely register the interventions, and it is a structural explanation rather than a claim of superiority: Table 2 shows the additive arm paying more total complexity for the same sixty rules.

The practical reading is about durability rather than quality. A structure that depends on prompting, review and metrics depends on the discipline of whoever is holding them, on every change, indefinitely. With an agent making the change sixty times over, the structure is whatever someone remembered to ask for. A structure that is a property of the composition holds when nobody is watching. Both architectures in this study reached a good placement under the right conditions. Only one of them reached it without being told.

That is as far as the evidence goes, and it is worth being blunt about how far that is not. Section 5.7 shows the placement result failing to predict any outcome a team would feel: the concentrated codebase was understood better by a cold reader in four conditions out of four, and was cheaper and faster to change in three of four. So the argument above cannot be completed in the usual way, by asserting that better placement pays off later. On this evidence it did not pay off within sixty change requests.

Two readings survive that, and this study cannot choose between them. The first is that sixty rules on one endpoint is too short a horizon: concentration compounds, and a study that stops before the concentrated arm becomes genuinely unworkable will see its costs before its benefits. The second is that the placement metrics are measuring something real but less important than the literature assumes, and that an architecture which distributes complexity buys structural stability by paying a comprehension and coordination cost that never amortises. The first is the hypothesis this line of work began with. The second is what this particular dataset, read on its own, supports slightly better. Distinguishing them needs a longer horizon or a human study, and neither is here.

7 Threats to validity

The statistic is exploratory and was not preregistered. S and R were defined after the four runs existed, so the intervals in Table 3 quantify sampling error and nothing else. They give no protection against a statistic chosen with the data already in view. S is also a range over four points, so it is sensitive to a single outlying condition by construction, and a range is less well behaved under the bootstrap than a mean because extrema are not smooth; the intervals are therefore indicative rather than exact. No multiple comparison correction is applied across the eleven placement metrics, although ten of the eleven have a bootstrap $P(R > 1)$ of 100% and would survive one. The metric set in Table 3 was chosen to span the concentration, change and coupling families rather than by a preregistered rule, and a different defensible selection would give different ratios. Three things are set against that. Every metric in the study is published, not only the reported ones (Section 8), and their full distribution is Figure 2. The three arrangement metrics that fail to separate the architectures are included. And the amount metrics are carried through the identical statistic as an internal control. The intervals establish that the asymmetry is not sampling noise. They do not make the study confirmatory, and it should still be read as an exploratory result that warrants a preregistered

replication.

R is unstable when its denominator is small. A ratio of two spreads inflates when the additive arm barely moves, which is precisely the condition the thesis predicts. The ratio is therefore reported beside both spreads rather than alone. Two metrics scoped to the entry handler illustrate the failure mode and are excluded from Table 3: `wmc_handler` has $S = 154\%$ and 135% , which reads as a tie, while the underlying move is 126.5 to 22.6 in one arm and 1.4 to 5.5 in the other.

Construct validity of “amount”. Two metrics published in the amount family do not conserve and are excluded from Table 1: `pmd_npath_total` and `halstead_effort` compose super linearly within a method and therefore fall when a method is split, which makes them arrangement sensitive. `total_fns` is retained but is the weakest member, since three of the four conditions were in effect asking for more functions. `halstead_vocab` is summed per file and so tracks file count rather than amount; it is not used.

The additive arm’s wiring is excluded from every measurement. OfficeFloor composes its endpoint in YAML, and the study counts YAML in a separate pool that is never folded into a Java denominator. That guard stops a wiring based architecture diluting any per line metric, but it cuts the other way too: the composition graph is exactly where the additive arm puts the structure that makes it additive, and no metric in this paper scores it. YAML has no cyclomatic complexity by construction. The numbers are that the additive arm’s fixed starting commit carries 460 lines of wiring, and the four conditions finish at 536 to 582, so the sixty rules added 76 to 122 lines that every amount metric records as zero, against 674 to 913 lines of Java added over the same rules. That is 9.4% to 11.8% of everything the additive arm added. The mutative arm has no wiring at all. The wiring is declarative and its growth is close to linear in the rule count, which is the reason for thinking it carries no hidden control flow, but a reader should know that roughly a tenth of what the additive arm added is outside the instruments, and that it is the tenth the architecture’s claim rests on.

The unit of analysis is one architecture per arm. Ten runs per cell is ten samples of the *agent*, not of the architecture. There is one Spring codebase and one OfficeFloor codebase, so every standard deviation in this paper describes model stochasticity and none of them describes variation between architectures of a kind. Re-running the agent more times would shrink those intervals without adding a single degree of freedom to a claim of the form “mutative architectures are plastic”. For that claim $n = 1$ in each arm, and the generalisation from these two codebases to their architectural styles is an argument, not a measurement.

The four conditions are not four draws from one population. S is a range over four interventions of deliberately different kinds and very different strengths: the gate fired on 195 of 1,200 checkpoints while the formula was in every prompt of its condition. The four are therefore not exchangeable, and a range over them is not a dispersion estimate in the usual sense. In practice the mutative arm’s range is set largely by the formula condition at one end and the control at the other, so $S(\text{SPRING})$ is closer to “the strongest available intervention moved it this far” than to “instructions move it this much on average”. The statistic is still the right comparison between arms, because both arms met the identical four interventions, but it should not be read as a variance.

Two reported metrics are weaker than the rest. `total_files` is close to definitional for an architecture that composes from many small units, so its R of 4.0 should not be read

as independent evidence. `mi_mean` was dropped from Table 3 during revision for the small denominator reason given above: its spreads are 5% and 1%, which is the same pathology as `wmc_handler` even though it produced a ratio favourable to the thesis.

A floor effect would explain this, and does not. The obvious alternative explanation is arithmetic rather than architectural: left alone, the additive arm already sits near an even distribution and so has little room to move, while the mutative arm sits far from one and so has a great deal. That account predicts that $S(\text{SPRING})$ tracks how far apart the two arms are when nothing is asked of them, which is their separation under the control condition. Across the eleven placement metrics the correlation between the two is $r = 0.262$, which is weak. `ccdist_file_gini` is the clearest counter-example: under the control the arms differ by only 6%, and the mutative arm still moves 34% under instruction while the additive arm moves 8%. Distance from the floor is therefore part of the story and not the whole of it.

One application, one endpoint, one model. All results are from a single application, a single endpoint, and `claude-opus-4-8`. Whether the plasticity gap holds for other frameworks, other domains or other models is open. The harness supports a model override for exactly this replication.

Conditions were not run concurrently. The four runs were executed on 2026-08-10 (control), 2026-09-01 (formula), 2026-09-03 (gated) and 2026-09-16 (cohesion), against the same pinned model identifier, which is recorded in each chain’s own configuration snapshot. The task plan and acceptance suite were frozen on 2026-08-08, before all four, so drift in the task is excluded. Drift in the model is not: a silent provider side change within that window cannot be ruled out from the data alone, and with one run per condition, condition and date are perfectly confounded.

The timing is asymmetric in a way that permits a partial check. The control ran three weeks before the others, while the three interventions ran within fifteen days of each other, so a provider side change between August and September would confound control against intervention but would leave the three interventions mutually comparable. Recomputing S and R over those three conditions alone, dropping the August control entirely, leaves the result intact: the eleven placement metrics give R from 2.9 to 10.6 with a median of 4.5, against 2.9 to 10.0 across all four, and the amount metrics stay at or below 1.1. The finding therefore does not rest on the run that is furthest away in time. It is a partial check and not a substitute for concurrent execution, which is the correct fix and which the harness supports.

Two conditions optimise a score defined by this study. The gated and formula conditions were built to move the structural impact score, so that score cannot be evidence for any claim about this experiment. It is excluded from every table here. Every metric that is used is either a published definition or a whole codebase count.

A reader who prefers erosion reaches the opposite conclusion. Section 5.4 reports that the borrowed whole-application erosion measure ranks the additive arm worse in three of four conditions. The argument given there is that erosion cannot see placement, so it is the wrong instrument for a claim about placement. That argument is not neutral: it is made by an author whose result improves when the contrary measure is set aside. A reader who holds that whole-application erosion is the better summary of degradation should read this paper as evidence against its own thesis, and the data to do that with is published.

Normalisation hides levels. Figure 1 normalises each architecture to its own control, which is what makes the two comparable on one axis, but it necessarily conceals the large absolute gap between them. Tables 1 and 4 carry the levels.

8 Reproducibility and data availability

Everything behind this paper is public, in two repositories.

The **harness** [14] holds the frozen change request plan, the acceptance suite, the measurement and analysis code, and the per checkpoint record for each run. Every metric reported here is recomputed by that analysis pass from committed source, rather than read back from any file an agent wrote.

The **run data** [17] holds the codebases themselves. Each chain is one branch:

evolve/<run>/<strategy>/<arm>/chain<n>

Every checkpoint on it is a commit carrying that checkpoint’s own raw capture: the agent transcript, the diff, the build log and the measurement record. The four runs reported here are 80 such branches, 10 chains for each of the two arms:

Condition	<run>	<strategy>
control	blind-202608100006	just-solve
cohesion prompt	blind-202609160027	cohesion-prompt
impact gated	blind-202609031757	impact_gated
formula provided	blind-202609010045	impact_gated

The last two share a strategy name because the formula condition predates the **advisory** enforcement mode; each chain’s own configuration snapshot, committed on its branch, is the authority on which is which.

Per metric figures for all eight series, across every metric in the study and not only those reported here, are produced by the analysis code, each with its definition and formula.

9 Conclusion

Across eight independent attempts at the same sixty change requests, the amount of complexity in the finished system varied by 7% on Halstead volume, and the amount the change requests added varied by at most 29% within an architecture. Against placement metrics that moved by 100% and more under the same interventions, that is a quantity refusing to be argued with. Tesler’s law and Brooks’ essential complexity survive the test.

What follows is that architecture is a question of arrangement, and that the durable property of an architecture is how little its arrangement depends on anything else. On that measure the two architectures here are not comparable. A prompt, a tool and a disclosed formula each moved the mutative architecture’s placement by between a third and a factor of two, and the best of them landed it where the additive architecture sits untouched. The same three interventions moved the additive architecture by 3% to 29%, because there was no slack for them to take up.

The claim is narrower than an endorsement, and deliberately so. Neither architecture was safe under coaching, since correctness fell in both. The additive arm paid more total complexity for the same rules. And the placement result predicted none of the outcomes this study measured: the concentrated codebase was understood better by a cold reader in every condition, and was cheaper and faster to change in three of four. Anyone reaching for these

results to argue that distributing complexity makes software cheaper to maintain will not find that here.

What is left is smaller, and still worth having. One of these two structures is negotiable and the other is not. Whether a structure is negotiable is a measurable property, it is not the same property as whether the structure is good, and it is worth knowing before an agent writes sixty changes into it.

Competing interests

The author is the creator and maintainer of OfficeFloor, which is one of the two architectures under test and the one the headline result favours. This is a material competing interest and it shapes how the paper should be read.

It extends past the framework to the codebases. The Spring arm is a fork of the community maintained Spring PetClinic REST application. The OfficeFloor arm was purpose-built for this comparison: an AI agent converted that same application, and the instruction it was given is published in full with the conversion [15]. Two things follow from that instruction, and they point in opposite directions.

The OfficeFloor baseline was therefore not hand-tuned by the author, which removes the most direct route by which an interested party could have shaped it. But the instruction asked for endpoints “decomposed into small single-responsibility functions (not controller-style methods)”. Decomposition is what OfficeFloor style means, so the instruction could hardly have said otherwise, yet it does mean the additive arm’s starting decomposition was specified rather than discovered. Readers should weigh the baseline accordingly. What the experiment measures is how each architecture behaves across sixty subsequent change requests, none of which mentions decomposition to either arm, but the two arms did not start from equally unprompted positions.

Its nature is worth stating precisely rather than leaving to inference. OfficeFloor is open source and has generated no revenue for the author, so there is no financial stake in the outcome. The interest that remains is intellectual and reputational, in a result that favours a design the author has spent years on. That is not a lesser kind of bias, and it is the kind this study is most exposed to, because the author chose which metrics to report.

One further interest belongs here for the same reason. The author distributes ImpactGate [16], an open source tool built on the structural impact score this study defines, so there is a standing interest in that score being regarded as sound. The score is excluded from every claim in this paper, for the reason given in Section 7: two of the four conditions were built to move it, so it cannot be evidence about this experiment. The tool is Apache licensed and has never been sold.

Three things are offered against it rather than as a dismissal of it. First, every input that could be tuned in OfficeFloor’s favour is fixed and public: the sixty change requests and their acceptance suite were frozen before any run reported here, the acceptance suite is black box and the agent never sees it, and both are in the repository. Second, the measurements are taken by third party tools on committed source, not by code written for this paper, and the analysis recomputes every figure from the commits rather than reading back anything an agent produced. Third, the results that go against the author’s interest are reported in the same detail as the results that favour it: the additive architecture added more total complexity than the mutative one in every condition (Table 2), its correctness fell under intervention just as the mutative arm’s did, three arrangement metrics fail to separate the architectures at all (Table 3), and every outcome measure the study records either fails to separate the arms or favours the mutative one (Table 5). That last result is the one most damaging to the author’s position, and it has a subsection of its own rather than a footnote.

None of that removes the interest. A reader who wants an independent check has the harness, the frozen plan and the four run identifiers in Section 8, and a replication needs no cooperation from the author.

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References

- [1] M. Aniche, “Java code metrics calculator (CK),” 2015. <https://github.com/mauricioaniche/ck>
- [2] F. P. Brooks Jr., “No Silver Bullet: Essence and Accidents of Software Engineering,” *IEEE Computer*, vol. 20, no. 4, pp. 10–19, 1987.
- [3] J. Chen, X. Xu, H. Wei, C. Chen and B. Zhao, “SWE-CI: Evaluating Agent Capabilities in Maintaining Codebases via Continuous Integration,” arXiv:2603.03823, 2026.
- [4] S. R. Chidamber and C. F. Kemerer, “A Metrics Suite for Object Oriented Design,” *IEEE Transactions on Software Engineering*, vol. 20, no. 6, pp. 476–493, 1994.
- [5] C. A. E. Goodhart, “Problems of Monetary Management: The U.K. Experience,” in *Papers in Monetary Economics*, Reserve Bank of Australia, 1975.
- [6] M. H. Halstead, *Elements of Software Science*. Elsevier, 1977.
- [7] A. E. Hassan, “Predicting Faults Using the Complexity of Code Changes,” in *Proc. 31st International Conference on Software Engineering (ICSE)*, pp. 78–88, 2009.
- [8] B. Henderson-Sellers, *Object-Oriented Metrics: Measures of Complexity*. Prentice Hall, 1996.
- [9] A. MacCormack, J. Rusnak and C. Y. Baldwin, “Exploring the Structure of Complex Software Designs: An Empirical Study of Open Source and Proprietary Code,” *Management Science*, vol. 52, no. 7, pp. 1015–1030, 2006.
- [10] T. J. McCabe, “A Complexity Measure,” *IEEE Transactions on Software Engineering*, vol. SE-2, no. 4, pp. 308–320, 1976.
- [11] D. A. Norman, *Living with Complexity*. MIT Press, 2010. Secondary discussion of the conservation of complexity.
- [12] P. Oman and J. Hagemester, “Metrics for Assessing a Software System’s Maintainability,” in *Proc. Conference on Software Maintenance (ICSM)*, pp. 337–344, 1992.
- [13] G. Orlanski, D. Roy, A. Yun, C. Shin, A. Gu, A. Ge, D. Adila, N. Roberts, F. Sala and A. Albarghouthi, “SlopCodeBench: Benchmarking How Coding Agents Degrade Over Long-Horizon Iterative Tasks,” arXiv:2603.24755, 2026.
- [14] D. Sagenschneider, “PetClinic-Evolve harness,” 2026. <https://github.com/officefloor/spring-petclinic-rest-long-degradation-test>

- [15] D. Sagenschneider, “Conversion to OfficeFloor YAML end point function injection,” 2026. The pull request carrying the OfficeFloor baseline, with the full conversion prompt in its description: <https://github.com/officefloor/spring-petclinic-rest/pull/3>
- [16] D. Sagenschneider, “ImpactGate: code review via the change impact measure,” 2026. Apache-2.0. <https://github.com/officefloor/ImpactGate>
- [17] D. Sagenschneider, “PetClinic-Evolve run data,” 2026. The per checkpoint commits and raw capture for every chain, as branches: <https://github.com/officefloor/spring-petclinic-rest>
- [18] D. Sagenschneider, “OfficeFloor: using office patterns to improve software design,” in *Proc. 18th European Conference on Pattern Languages of Programs (EuroPLoP)*, ACM, pp. 1–27, 2013. <https://doi.org/10.1145/2739011.2739013>
- [19] Spring Framework, VMware. <https://spring.io/projects/spring-framework>
- [20] M. Strathern, “‘Improving ratings’: audit in the British University system,” *European Review*, vol. 5, no. 3, pp. 305–321, 1997.
- [21] L. Tesler, “The Law of Conservation of Complexity,” c. 1984. <http://www.nomodes.com/larry-tesler-consulting/complexity-law> The law has no journal publication; this is the author’s own statement of it.